The radion of 7+1 spacetime: symbols and their definitions

A dictionary of every symbol used in the document on the radion, with its defining equation and its name in the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

For each symbol this dictionary gives the mathematical notation, the name used in the Mathematica notebook 7+1.nb (where there is one; the names of section 26 begin with `rd`; "—" means that the notebook has no separate name), the defining equation and its meaning, grouped by topic in the order of the document. Longer defining equations follow each table as numbered displayed formulas. Section 11 lists the symbols that look alike, section 12 every Mathematica name of section 26. Every equation needed to define a symbol is written in this document.

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1. Conventions

- Units: $\hbar = c = 1$; masses and inverse lengths in GeV; $\hbar c = 1.973269804 \times 10^{-16}$ GeV m (Mathematica `tbHbarCm`).
- Coordinates $X = (x_0, x_1, \ldots, x_7)$ (Mathematica `X = {x0, …, x7}`): x_0 is time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ the four extra space-like coordinates: on the 4-torus periodic with period L_c, on the 4-sphere the angles $x_4, x_5, x_6 \in [0, \pi]$, $x_7 \in [0, 2\pi)$.
- Indices: $A, B = 0, \ldots, 7$; $\mu, \nu = 0, \ldots, 3$ (ordinary spacetime, the directions of the world brane); $i, j = 4, \ldots, 7$ (the extra space); $a, b = 1, \ldots, 4$ (frame indices of the extra space, or the coordinates X_a of $\mathbb{R}^5$ when $a, b \le 5$). A prime is d/dx_0 (cosmology) or d/dy (warped metrics); ln is the natural logarithm.
- Curvature (mostly-minus signature): $R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho{}_{\nu\sigma} - \partial_\nu \Gamma^\rho{}_{\mu\sigma} + \Gamma^\rho{}_{\mu\lambda}\Gamma^\lambda{}_{\nu\sigma} - \Gamma^\rho{}_{\nu\lambda}\Gamma^\lambda{}_{\mu\sigma}$, $R_{\sigma\nu} = R^\rho{}_{\sigma\rho\nu}$, $R = g^{\sigma\nu} R_{\sigma\nu}$, $G^A{}_B = R^A{}_B - \frac{1}{2}\delta^A_B R$; field equations $G^A{}_B - \Lambda \delta^A_B = \kappa T^A{}_B$.

The signature: $\{7,1,0\}$ and $\mathrm{diag}(+1,-1,\ldots,-1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7,1,0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1,+1,\ldots,+1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like = $+1$, time-like = -1). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1,-1,\ldots,-1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1,\ldots,7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p,q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1,7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta = \det\eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7,1,0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$.

2. Coordinates, metrics and the two extra spaces

symbol	Mathematica	definition	meaning
x, y	`{x0, …, x3}`, `{x4, …, x7}`	$x = (x_0,\ldots,x_3)$, $y = (x_4,\ldots,x_7)$	a point of ordinary spacetime; a point of the extra space
G_{AB}, ds^2	—	(2.1)	the eight-dimensional metric of the radion ansatz
$g_{E\mu\nu}$	`{rdLp^2, -rdSa^2, …}` (homogeneous)	(2.1)	the four-dimensional Einstein-frame metric
N, a	`rdLp[x0]`, `rdSa[x0]`	$g_E = \mathrm{diag}(N^2, -a^2, -a^2, -a^2)$	lapse and scale factor of the homogeneous checks
γ_{ij}	`rdB0^2 {1,1,1,1}`, `rdR0^2 rdGam4`	$b_0^2\delta_{ij}$ (T^4); $r_0^2\hat\gamma_{ij}$ (S^4)	the fixed reference metric of the extra space
$\hat\gamma$	`rdGam4`	(2.2)	the metric of the unit 4-sphere
T^4	—	$\mathbb{R}^4/(L_c\mathbb{Z})^4$	the 4-torus (the auxiliary space of 7+1.nb)
S^4	—	(2.3)	the 4-sphere (for comparison)
L_c	`rdLc`	coordinate period	period of each torus coordinate
b, b_0	`b4[x0]` (section 11), `rdB0`	$b = b_0 e^{\beta}$	size of the torus; its reference value
r, r_0	`rdR0` (r_0)	$r = r_0 e^{\beta}$	radius of the sphere; its reference value
V, V_0	`v0` (argument of `rdMini`)	(2.4)	proper volume of the extra space; at $\beta = 0$
$V_\perp$, $\ell_\perp$	`rIIVp`, `tbLp`	$V_\perp = M_{\rm Pl}^2/M_8^6$, $\ell_\perp = V_\perp^{1/4}$	volume of the torus today; its proper period
r_{S^4}	`rdRS4`	$(3V_\perp/(8\pi^2))^{1/4}$	radius of a sphere with the volume $V_\perp$
X_a	`rdEmb`	(2.3)	embedding of S^4 in $\mathbb{R}^5$, $a = 1,\ldots,5$
$R_{\rm int}$	`rint` (argument of `rdMini`)	0 (T^4), $12/r^2$ (S^4)	scalar curvature of the extra space itself (positive for the sphere)
R, R_E	`rdScal[g, X]`	section 1	scalar curvature of G; of g_E
R_{ij}	`rdRic`, `rdRicS4`	section 1	Ricci tensor

$$ds^2 = G_{AB}dX^A dX^B = e^{-4\beta(x)} g_{E\mu\nu}(x)dx^\mu dx^\nu - e^{2\beta(x)}\gamma_{ij}(y)dy^i dy^j, \tag{2.1}$$

$$\hat{\gamma} = \mathrm{diag}\big(1,\ \sin^2 x_4,\ \sin^2 x_4 \sin^2 x_5,\ \sin^2 x_4 \sin^2 x_5 \sin^2 x_6\big), \tag{2.2}$$

$$X_a = r\left(\sin x_4 \sin x_5 \sin x_6 \cos x_7,\ \sin x_4 \sin x_5 \sin x_6 \sin x_7,\ \sin x_4 \sin x_5 \cos x_6,\ \sin x_4 \cos x_5,\ \cos x_4\right), \tag{2.3}$$

$$V = \int d^4y \sqrt{\det(e^{2\beta}\gamma)} = V_0 e^{4\beta}, \qquad V_0 = b_0^4 L_c^4\ (T^4), \quad V_0 = \frac{8\pi^2}{3} r_0^4\ (S^4). \tag{2.4}$$

3. The radion, the Einstein frame and the reduced action

symbol	Mathematica	definition	meaning
β	`rdBt[x0]` (field), `rdBe` (number)	$\ln(b/b_0)$ or $\ln(r/r_0)$	the radion (dimensionless)
ϕ	—	(3.1)	the canonical radion (mass dimension 1)
α, β (the note)	—	$ds^2 = e^{2\alpha\phi} g + e^{2\beta\phi}\gamma$	the constants of the note's ansatz; $\beta\phi$ of the note is our $\beta(x)$
β_n	—	(3.3)	the constant β for n extra dimensions
φ	`rdPh[x0]`	$\beta = \beta_n \varphi$	the radion in the normalization of the note (dimensionless)
n	`n` in `rdKinOK`, `rdKinC`	$D = 4 + n$	number of extra dimensions
c_n	`rdKinC[n]`, `rdKinCs`	(3.2)	coefficient of the radion kinetic term
κ	`rdKap`	$8\pi G_8 = M_8^{-6}$	eight-dimensional gravitational coupling
M_8	`rIIM8`	10 TeV	eight-dimensional Planck scale of the examples
$M_{\rm Pl}$	`rIIMPl`, `rdMP2` ($M_{\rm Pl}^2$)	(3.1)	reduced four-dimensional Planck mass, 2.435×10^{18} GeV
G_4, G_8	—	$8\pi G_4 = M_{\rm Pl}^{-2}$, $8\pi G_8 = \kappa$	Newton constants
Λ	`rdLam`	section 1	bulk cosmological constant ($\Lambda > 0$: de Sitter)
U, U_0, U_1, K_U	`rdU`, `rdU0`, `rdU1`, `rdKU`, `rdUq`	(4.1)	stabilizing energy density of the bulk and its expansion
S	—	(3.4)	the action
Ω	`rdOmg`, `rdOmE`	$h = \Omega^2 \eta$	conformal factor (section 6)

$$\phi = \sqrt{12}\, M_{\rm Pl}\, \beta, \qquad M_{\rm Pl}^2 = \frac{V_0}{\kappa}, \qquad 6M_{\rm Pl}^2(\partial\beta)^2 = \frac{1}{2}(\partial\phi)^2, \tag{3.1}$$

$$-\frac{1}{2\kappa}\int d^n y \sqrt{|G|} R = \sqrt{|g_E|}\Big[-\frac{M_{\rm Pl}^2}{2} R_E + \frac{M_{\rm Pl}^2}{2} c_n (\partial\beta)^2\Big] + (\text{total derivative}), \tag{3.2}$$

$$c_n = \frac{n(n+2)}{2},$$

$$2\alpha + n\beta_n = 0; \qquad \text{the note: } \beta_n = \sqrt{\frac{2}{n(n+2)}}\ \ (c_n\beta_n^2 = 1); \tag{3.3}$$

$$\text{canonical: } \beta_n = \frac{1}{\sqrt{n(n+2)}}\ \ (c_n\beta_n^2 = \tfrac{1}{2}),$$

$$S = -\frac{1}{2\kappa}\int d^8X \sqrt{|G|}(R + 2\Lambda) - \int d^8X \sqrt{|G|}\, U + S_{\rm brane} + S_{\rm matter}. \tag{3.4}$$

4. The potential and the mass

symbol	Mathematica	definition	meaning
$V(\beta)$	`rdV[curv, flux]`	(4.2)	Einstein-frame potential of the radion
σ	`curv` (argument of `rdV`)	0 (T^4), 1 (S^4)	switch of the curvature term
F, ρ_F, q	`rdFx`, `rdRhoF`, `rdQ2` (q^2)	(4.3)	flux term; flux energy density; flux quantum (illustration)
$W(a,b)$	argument of `feStress`	(4.3)	comoving energy of a source
ε, p_a, p_b	`rdEps`, `rdPa`, `rdPb`	(4.3)	energy density, pressures along ordinary and extra space
H_a, H_b	`Ha`, `Hb` (section 20)	a'/a, b'/b	expansion rates; $H_b = \beta'$
m_r, m_r^2	`rdM2L` (Λ-only case)	(4.4)	mass of the radion (10 GeV in the examples of 7+1.nb)
H	`rdHE2` (H^2)	$H^2 = V(0)/(3M_{\rm Pl}^2)$	Hubble rate of four-dimensional de Sitter space
Φ, m_8	`rdPhi[x4, x5]`, `rdMm`	(4.5)	a bulk scalar and its mass (the note's stabilizer)

$$U(\beta) = U_0 + U_1\beta + \frac{K_U}{2}\beta^2, \tag{4.1}$$

$$V(\beta) = M_{\rm Pl}^2\Big[e^{-4\beta}\big(\Lambda + \kappa U(\beta)\big) - \frac{6\sigma}{r_0^2}e^{-6\beta} + F\,e^{-12\beta}\Big], \tag{4.2}$$

$$\varepsilon = \frac{W}{a^3b^4}, \quad p_a = -\frac{\partial_a W}{3a^2b^4}, \quad p_b = -\frac{\partial_b W}{4a^3b^3}; \qquad W = \frac{a^3b^4q^2}{2b^8} : \ \rho_F = \frac{q^2}{2b^8}, \ F = \kappa\rho_F(b_0), \tag{4.3}$$

$$m_r^2 = \frac{d^2V}{d\phi^2}\Big|_0 = \frac{V''(0)}{12M_{\rm Pl}^2}, \tag{4.4}$$

$$\sqrt{|G|}\Big(\frac{1}{2}G^{AB}\partial_A\Phi\partial_B\Phi - \frac{1}{2}m_8^2\Phi^2\Big) = -\sqrt{|g_E|}\sqrt{\gamma}\,e^{-4\beta}\Big(\frac{1}{2}e^{-2\beta}|\nabla\Phi|_\gamma^2 + \frac{1}{2}m_8^2\Phi^2\Big). \tag{4.5}$$

5. The radion and the world brane

symbol	Mathematica	definition	meaning
W (brane)	—	$y = 0$ (T^4), $x_4 = 0$ (S^4)	the world brane
$h_{\mu\nu}$	—	(5.1)	induced metric on the brane
$T^\mu{}_\mu$	—	trace of $T^{\mu\nu}$	trace of the energy–momentum tensor of brane matter
M, M'	`rdMs`	masses	point masses on the brane
g	—	(5.2)	coupling of the radion to $T^\mu{}_\mu$
α (fifth force)	—	(5.2)	strength of the radion force relative to gravity
λ (Yukawa)	—	$1/m_r$	range of a Yukawa force
r (distance)	—	$\lvert\vec{x} - \vec{x}'\rvert$	distance between two masses

$$h_{\mu\nu} = e^{-4\beta}g_{E\mu\nu} \quad (n = 4;\ \text{in general } e^{-n\beta}g_{E\mu\nu}), \tag{5.1}$$

$$\mathcal{L}_{\rm int} = g\,\phi\,T^\mu{}_\mu, \quad g = \frac{1}{\sqrt{3}M_{\rm Pl}}\ \Big(\text{in general } \frac{1}{M_{\rm Pl}}\sqrt{\frac{n}{2(n+2)}}\Big), \quad \alpha = 2(gM_{\rm Pl})^2 = \frac{n}{n+2}. \tag{5.2}$$

6. The Higgs boson and the radion

symbol	Mathematica	definition	meaning
H, φ_H	`rdPhE` (φ_H)	$H^\dagger H = \varphi_H^2/2$, $\varphi_H = v + h$	Higgs doublet; its radial field
v	`vEW`, `rdVv`	$(\sqrt{2}G_F)^{-1/2} = 246.22$ GeV	Higgs vacuum value
h, r	`rdHh[x0, x1]`, `rdRr[x0, x1]`	section 6	bare Higgs and radion fields (canonical r)
λ (Higgs)	`rdLamH`	$V = \frac{\lambda}{4}(\varphi_H^2 - v^2)^2$	Higgs quartic coupling
ξ	`rdXi`	(6.1)	curvature coupling of the Higgs ($\frac{1}{6}$: conformal)
Λ_r, γ	`rdO1` ($1/\Lambda_r$), `rdGam`	$\Omega = 1 + r/\Lambda_r + \ldots$, $\gamma = v/\Lambda_r$	coupling scale of the radion; mixing parameter
m_{h0}, m_{r0}	`rdMh0`, `rdMr0`	$m_{h0}^2 = 2\lambda v^2$	bare Higgs and radion masses
c	`rdc4`	(6.2)	coefficient of $\Omega\Box\Omega$
K, K_{hr}, K_{rr}	`rdKm`, `rdKhr`, `rdKrr`	(6.3)	kinetic matrix of (h, r)
S (shift), Z	`rdSm`, `rdZ`, `rdZ2` (Z^2)	(6.4)	kinetic shift; normalization
M'	`rdMp`	(6.5)	mass matrix of (h', r')
$Q(\theta)$, θ	`rdQ[th]`, `rdth`	(6.5)	rotation; mixing angle
h_m, r_m	—	(6.5)	mass eigenstates
C_h, C_r	—	(6.5)	coefficients of h_m in h and r
$b_{\rm EM}$, $F_{\rm SM}$, $b_{\rm QCD}$, $F_{\rm heavy}$	—	$-\frac{11}{3}$, -6.5, 7, $-\frac{4}{3}$ (the note)	trace-anomaly and loop coefficients
Γ	`rdHRnum`, `rdHR71`	(6.6)	decay rates of the Higgs boson

$$\mathcal{L} = \sqrt{|h|}\Big[\frac{1}{2}h^{\mu\nu}\partial_\mu\varphi_H\partial_\nu\varphi_H - \frac{\lambda}{4}(\varphi_H^2 - v^2)^2 + \frac{\xi}{2}R(h)\varphi_H^2\Big] + \frac{1}{2}(\partial r)^2 - \frac{1}{2}m_{r0}^2 r^2, \tag{6.1}$$

$$h = \Omega^2\eta: \quad \sqrt{|h|}R(h) = c\,\Omega\Box\Omega, \quad c = -6, \quad \Box = \partial_0^2 - \nabla^2, \tag{6.2}$$

$$\mathcal{L}^{(2)} = \frac{1}{2}\begin{pmatrix}\partial h & \partial r\end{pmatrix} K \begin{pmatrix}\partial h \\ \partial r\end{pmatrix} - \frac{1}{2}m_{h0}^2 h^2 - \frac{1}{2}m_{r0}^2 r^2, \quad K = \begin{pmatrix}1 & 6\xi\gamma \\ 6\xi\gamma & 1 + 6\xi\gamma^2\end{pmatrix}, \tag{6.3}$$

$$\begin{pmatrix}h \\ r\end{pmatrix} = S\begin{pmatrix}h' \\ r'\end{pmatrix}, \quad S = \begin{pmatrix}1 & -6\xi\gamma/Z \\ 0 & 1/Z\end{pmatrix}, \quad S^{\mathsf{T}}KS = 1 \iff Z^2 = 1 + 6\xi\gamma^2(1 - 6\xi), \tag{6.4}$$

$$M' = S^{\mathsf{T}}\mathrm{diag}(m_{h0}^2, m_{r0}^2)S, \quad \begin{pmatrix}h' \\ r'\end{pmatrix} = Q(\theta)\begin{pmatrix}h_m \\ r_m\end{pmatrix}, \quad Q(\theta) = \begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}, \tag{6.5}$$

$$\tan 2\theta = \frac{12\gamma\xi Z m_{h0}^2}{m_{r0}^2 - m_{h0}^2(Z^2 - 36\xi^2\gamma^2)}, \quad C_h = \cos\theta - \frac{6\xi\gamma}{Z}\sin\theta, \quad C_r = \frac{\sin\theta}{Z},$$

$$\frac{\Gamma(h \to \gamma\gamma)}{\Gamma_{\rm SM}} = \Big|C_h + C_r\gamma\frac{b_{\rm EM}}{F_{\rm SM}}\Big|^2, \qquad \frac{\Gamma(h \to gg)}{\Gamma_{\rm SM}} = \Big|C_h + C_r\gamma\frac{b_{\rm QCD}}{F_{\rm heavy}}\Big|^2. \tag{6.6}$$

7. Kaluza–Klein levels on T^4 and S^4

symbol	Mathematica	definition	meaning
Ψ, ψ_n, Y_n	—	$\Psi = \sum_n \psi_n(x)Y_n(y)$	a bulk field, its four-dimensional modes and the eigenfunctions
Δ_γ	—	Laplacian of γ	Laplacian of the extra space
μ, μ_n	—	$-\Delta_\gamma Y_n = \mu_n^2 Y_n$	Kaluza–Klein mass
$\vec{n}$, k	—	$\vec{n} \in \mathbb{Z}^4$, $k = \lvert\vec{n}\rvert^2$	torus mode number; shell
$r_4(k)$, $\sigma(k)$	`rIIr4`, `rIIsig`	(7.1)	number of $\vec{n}$ with $\lvert\vec{n}\rvert^2 = k$; divisor sum
l, d_l	`rdDl[l]`	(7.2)	degree of a spherical harmonic; multiplicity
$N(\mu)$, x, L, y	`rdNS4[x]`, `rdWeyl`	(7.3)	number of states below μ; $x = \mu r$; largest l; $y = L + 2$

symbol	Mathematica	definition	meaning
Z_l, $C_l^{(3/2)}$	`rdZon[l]`, `GegenbauerC`	(7.4)	zonal harmonic; Gegenbauer polynomial
L_{ab}	—	$X_a\partial_b - X_b\partial_a$	rotations of $\mathbb{R}^5$ (Killing vectors of S^4)

$$r_4(k) = 8\sigma(k) - 32\sigma(k/4), \quad \sigma(k) = \sum_{d|k} d \ (\sigma(k/4) = 0 \text{ unless } 4|k); \quad T^4: \ \mu = \frac{2\pi|\vec{n}|}{\ell_\perp}, \tag{7.1}$$

$$S^4: \ \mu = \frac{\sqrt{l(l+3)}}{r}, \quad d_l = \frac{(2l+3)(l+1)(l+2)}{6} = \binom{l+4}{4} - \binom{l+2}{4}, \tag{7.2}$$

$$N(\mu) \simeq \frac{V\mu^4}{32\pi^2}; \quad S^4: \ N = \sum_{l\le L} d_l = \frac{y^2(y^2-1)}{12}, \quad L(L+3) \le x^2 < (L+1)(L+4), \tag{7.3}$$

$$Z_l(x_4) = \sqrt{\frac{d_l}{V}}\,\frac{C_l^{(3/2)}(\cos x_4)}{C_l^{(3/2)}(1)}, \qquad C_l^{(3/2)}(1) = \frac{(l+1)(l+2)}{2}. \tag{7.4}$$

8. Spinors on S^4 and dirac16complex

symbol	Mathematica	definition	meaning
ρ_k	`rhoK` (section 17.2)	$\rho_k\rho_l + \rho_l\rho_k = -2\delta_{kl}$	the 4×4 matrices of the transverse spinor index
$\tilde\gamma_a$	`rdGt`	$i\rho_{a+3}$	Hermitian Dirac matrices of the extra space
e^a, $e_i{}^a$	`rdHf`	(8.1)	orthonormal frame of S^4
ω	`rdOm`	spin connection	connection coefficients of the frame
∇_i	`rdNab`	(8.1)	spinor covariant derivative
$\not{D}$	—	$\tilde\gamma^a e_a{}^i\nabla_i$	Dirac operator of the extra space
ε_s, ε_0, s	`rdKS[s]`	(8.2)	Killing spinors; a constant spinor; $s=\pm1$
λ (Dirac)	—	(8.3)	eigenvalue of $\not{D}$
m	—	mass parameter	eight-dimensional mass of dirac16complex

$$e^a = (r,\ r\sin x_4,\ r\sin x_4\sin x_5,\ r\sin x_4\sin x_5\sin x_6)\,dx^{a+3}, \qquad \nabla_i = \partial_i + \frac{1}{4}\omega_{iab}\tilde\gamma^a\tilde\gamma^b, \tag{8.1}$$

$$\varepsilon_s = e^{s\frac{i}{2}x_4\tilde\gamma_1}e^{\frac{1}{2}x_5\tilde\gamma_1\tilde\gamma_2}e^{\frac{1}{2}x_6\tilde\gamma_2\tilde\gamma_3}e^{\frac{1}{2}x_7\tilde\gamma_3\tilde\gamma_4}\varepsilon_0, \quad \nabla_i\varepsilon_s = s\frac{i}{2r}e_i{}^a\tilde\gamma_a\varepsilon_s, \quad \not{D}\varepsilon_s = s\frac{2i}{r}\varepsilon_s, \tag{8.2}$$

$$\lambda = \pm\frac{i(l+2)}{r} \ (\text{degeneracy } 4\tbinom{l+3}{3} \text{ each}), \qquad |\lambda|^2 \ge \frac{nR_{\rm int}}{4(n-1)} = \frac{4}{r^2}, \qquad m_{\rm KK} = \sqrt{m^2+|\lambda|^2}. \tag{8.3}$$

9. The warped radion (Randall–Sundrum)

symbol	Mathematica	definition	meaning
$A(y)$, k	`rdA[rdy]`, `rdk`	(9.1)	warp function; AdS curvature scale
y, R	`rdy`, `rdRR`	$y \in [-\pi R, \pi R]$, $y \simeq -y$	the fifth coordinate; the radius
$T(x)$	`rdT[rdt]`, `rdTs`	(9.2)	the radius as a field (geometric radion)
μ, f	`rdmu`, `rdfm`	(9.2)	canonical warped radion; its normalization
$f_{\rm UV}$, $f_{\rm IR}$, f	`rdfUV`, `rdfIR`, `rdf`	(9.1)	brane tensions
λ_A, λ_B, $\lambda_{\rm UV}$, $\lambda_{\rm IR}$	—	the note's tensions	the same tensions in the note
$K^\mu{}_\nu$, $S_{\mu\nu}$	`rdKmix`, `rdKjump`	(9.1)	extrinsic curvature; brane energy–momentum
M, M_5, κ_8	—	table of section 9.2 of the document	Planck scales of other normalizations
$\Phi(y)$, m, ν, ϵ	`rdPhy`, `rdMg`, `rdNu`, `rdeps`	(9.3)	Goldberger–Wise scalar, its mass, exponent, small parameter
v_A, v_B, C_1, C_2	`rdvA`, `rdvB`, `rdC1`, `rdC2`	(9.3)	brane values of Φ; integration constants
$\langle\mu\rangle$, $\langle T\rangle$	`rdmu0`	(9.4)	minimum of $V(\mu)$
c_5, $[A']$	`rdc5`, `rdJmp`	3; $2k$	coefficient of A'' in $G^\mu{}_\nu$; jump of A' at the UV brane

$$ds^2 = e^{-2A(y)}\eta_{\mu\nu}dx^\mu dx^\nu - dy^2, \quad G^\mu{}_\nu = 3A'' - 6A'^2, \quad G^y{}_y = -6A'^2, \tag{9.1}$$
$$f_{\rm UV} = -f_{\rm IR} = \frac{6k}{\kappa}, \qquad K^\mu{}_\nu = \frac{1}{2}g^{\mu\alpha}\partial_y g_{\alpha\nu},$$

$$ds^2 = e^{-2kT(x)|y|}g_{\mu\nu}dx^\mu dx^\nu - T(x)^2 dy^2, \quad \mu = f\,e^{-k\pi T}, \quad f^2 = \frac{6}{\kappa k}, \quad M_{\rm Pl}^2 = \frac{1-e^{-2\pi kR}}{\kappa k}, \tag{9.2}$$

$$\Phi'' - 4k\Phi' - m^2\Phi = 0, \quad \Phi = e^{2ky}(C_1 e^{\nu k y} + C_2 e^{-\nu k y}), \quad \nu = \sqrt{4 + m^2/k^2}, \quad \epsilon = \frac{m^2}{4k^2}, \tag{9.3}$$

$$V(\mu) = 4k\Big(\frac{\mu}{f}\Big)^4\Big[v_B - v_A\Big(\frac{\mu}{f}\Big)^\epsilon\Big]^2, \quad \Big(\frac{\langle\mu\rangle}{f}\Big)^\epsilon = \frac{v_B}{v_A}, \quad m_r^2 = \frac{4}{3}\kappa k^2\epsilon^2 v_B^2\Big(\frac{\langle\mu\rangle}{f}\Big)^2. \tag{9.4}$$

10. Numbers and constants

symbol	Mathematica	value	source
$M_{\rm Pl}$	`rIIMPl`	2.435×10^{18} GeV	reduced Planck mass $(8\pi G_N)^{-1/2}$
M_8	`rIIM8`	10^4 GeV	choice of sections 22–24
m_r	—	10 GeV	choice of sections 22–24
m_h, m_{h0}	`mHb`	125.20 GeV	Particle Data Group 2025
G_F	`gFermi`	1.1663788×10^{-5} GeV^{-2}	Particle Data Group
$\hbar c$	`tbHbarCm`	$1.973269804 \times 10^{-16}$ GeV m	CODATA 2018
38.6 μm	—	Eöt-Wash range bound for $\lvert\alpha\rvert = 1$	Lee et al., Phys. Rev. Lett. 124 (2020) 101101
$\langle\mu\rangle$	—	3 TeV	the note's warped example

11. Symbols that look alike

- β: the radion field $\beta(x)$ (this document) and the constant β of the note's ansatz; β_n the constant for n extra dimensions. Our $\beta(x)$ equals the note's product $\beta\phi(x)$.
- α: the note's constant in $e^{2\alpha\phi}$ and the strength α of a Yukawa (fifth) force.
- ϕ, φ, φ_H, Φ: the canonical radion ϕ; the note's dimensionless radion φ; the radial Higgs field φ_H; a bulk scalar Φ.
- λ: the Higgs quartic coupling, the Dirac eigenvalue on S^4, the range of a Yukawa force, and (with subscripts) the note's brane tensions.
- ϵ, ε: the Goldberger–Wise parameter $\epsilon = m^2/(4k^2)$; the energy density ε; the Killing spinors ε_s.
- γ: the reference metric γ_{ij} ($\hat{\gamma}$ for the unit sphere), the Higgs–radion parameter $\gamma = v/\Lambda_r$, and the photon in $h \to \gamma\gamma$; $\tilde{\gamma}_a$ are Dirac matrices.
- σ: the curvature switch of $V(\beta)$ and the divisor sum $\sigma(k)$.
- r: the radius of the sphere, the distance between two masses, and the bare radion field $r(x)$; r_m the radion mass eigenstate; $r_4(k)$ the lattice multiplicity.
- h: the induced metric $h_{\mu\nu}$, the bare Higgs field $h(x)$ and the mass eigenstate h_m.
- R: scalar curvature (R, R_E, R_{int}), the Randall–Sundrum radius R, and $R(h)$, the curvature of the induced metric.
- K: the stiffness K_U, the kinetic matrix K, the extrinsic curvature $K^\mu{}_\nu$.
- f: the brane tension f (f_{UV}, f_{IR}), the normalization f of the warped radion, and f_W (the tension of the world brane of 7+1).
- Λ, Λ_r: the bulk cosmological constant; the coupling scale of the radion.
- M: a point mass, the Randall–Sundrum scale M, M_5, M_8, M_{Pl}, and the mass matrix M'.
- S: the action S, the kinetic shift matrix S, the brane energy–momentum $S_{\mu\nu}$, the sphere S^4.
- x: the coordinates and the dimensionless $x = \mu r$ of Weyl's law; y: the extra coordinates, the fifth coordinate of RS1, and $y = L + 2$.
- Z: the normalization Z and the zonal harmonics Z_l; $\mathbb{Z}$ the integers.
- c: the coefficient $c = -6$ and the coefficients c_n; C_1, C_2, C_h, C_r, $C_l^{(3/2)}$ are different objects.

12. Mathematica names

12.1 The names of section 26

name	meaning
`rdChr`, `rdRic`, `rdScal`, `rdEin`, `rdEL`, `rdEL2`	Christoffel symbols, Ricci tensor, scalar curvature, mixed Einstein tensor of a diagonal metric; Euler–Lagrange operators in x_0 and in (x_0, x_1)
`rdGam4`, `rdRicS4`, `rdR0`, `rdB0`, `rdLc`	$\hat{\gamma}$; the Ricci tensor of S^4; r_0; b_0; L_c
`rdMini[gam, rint, v0]`	the reduction test of section 26.1 (homogeneous metrics; Euler–Lagrange expressions for N, a, β)
`rdLp`, `rdSa`, `rdBt`, `rdPh`	$N(x_0)$, $a(x_0)$, $\beta(x_0)$, $\varphi(x_0)$
`rdKap`, `rdLam`, `rdU`, `rdU0`, `rdU1`, `rdKU`, `rdUq`, `rdMP2`	κ, Λ, U, U_0, U_1, K_U, $U(\beta)$, M_{Pl}^2
`rdKinOK`, `rdKinC`, `rdKinCs`, `rdY`	the kinetic-term test; the solved coefficient c_n; $(c_1, \ldots, c_4)$; extra coordinates of T^n
`rdES`, `rdET`, `rdE00`, `rdE11`, `rdE44`, `rdb`	Einstein tensors of the sphere and torus cosmologies; their components; the symbol b

name	meaning
rdSolS, rdRadS, rdEps, rdPa, rdPb	the solution for H_a', H_b'; the radion equation on S^4; ε, p_a, p_b
rdBe, rdV, rdRHS	β (number); $V(\beta)$; the right-hand side of the radion equation
rdSolT, rdSolS2, rdUb, rdR2	the conditions on T^4 and on S^4; U at the solution; the radion equation there
rdVL, rdLamL, rdHE2, rdM2L, rdDS	the Λ-only sphere: V, Λ, H^2, m^2, the field equations of $dS_4 \times S^4$
rdVF, rdFx, rdSolF, rdStF, rdQ2, rdRhoF	the flux sphere: V, F, the solution, the stress of W, q^2, ρ_F
rdGdS, rdSqG, rdPhi, rdMm	metric and $\sqrt{\lvert G\rvert}$ of the Goldberger–Wise check; $\Phi(x_4, x_5)$; m_8
rdPP, rdMs, rdNn, rdEW	the expansion of $-Me^{-n\beta/2}$; M; n; the Eöt-Wash numbers
rdDl, rdZon, rdNS4, rdWeyl	d_l; $C_l^{(3/2)}(\cos x_4)$; the exact count on S^4; Weyl deviations
rdEmb, rdJe	the embedding X_a and its Jacobian
rdGt, rdHf, rdOm, rdNab, rdKS, rdZeroQ, rdv4 … rdv7	$\tilde{\gamma}_a$; frame lengths; connection; ∇_i; Killing spinors; the exact zero test (half angles)
rdSpW, rdRS4, rdToday	spinor Weyl ratio; r_{S^4}; the numbers today
rdXs5, rdt, rdx1 … rdx3, rdy, rdz1 … rdz3	coordinates of the five- and eight-dimensional warped metrics
rdG5, rdA, rdc5, rdJmp, rdk, rdf, rdfUV, rdfIR	RS1 Einstein tensor; $A(y)$; 3; $2k$; k; a tension; f_{UV}, f_{IR}
rdKmix, rdKjump, rdRR	$K^{\mu}{}_{\nu}$; its jump at the UV brane; R
rdR5T, rdT, rdTs, rdT1, rdT2, rdLb, rdC2, rdL1	the radion reduction of RS1: curvature, $T(x_0)$, T, T', T'', Lagrangian, coefficients
rdG8w, rdA1, rdA2	eight-dimensional warped Einstein tensor; A', A'' as unknowns
rdGWL, rdPhy, rdMg, rdGWsol, rdC1, rdNu	Goldberger–Wise Lagrangian; $\Phi(y)$; m; the solution; C_1; ν
rdVgw, rdmu, rdfm, rdvA, rdvB, rdeps, rdmu0, rdV2gw, rdMgw, rdMgwNote	$V(\mu)$, μ, f, v_A, v_B, ϵ, $\langle\mu\rangle$, $V''(\langle\mu\rangle)$, the mass coefficient with $f^2 = 6/(\kappa k)$ and with $12/(\kappa k)$
rdc4, rdOmg, rdOmE, rdO1, rdO2, rdEp	c; $\Omega(x_0, x_1)$; $\Omega = 1 + o_1\epsilon r + o_2\epsilon^2 r^2$; $1/\Lambda_r$; o_2; bookkeeping ϵ
rdRr, rdHh, rdPhE, rdVv, rdRh, rdLbr, rdLamH, rdXi	r, h, φ_H, v, $R(h)$, the brane Lagrangian, λ, ξ
rdMr0, rdMh0, rdHL1, rdHL2, rdGam, rdLq	m_{r0}, m_{h0}; first- and second-order Lagrangians; γ; the quadratic Lagrangian
rdKhr, rdKrr, rdKm, rdKnote, rdSm, rdZ, rdZ2	kinetic coefficients and matrix; the note's literal matrix; S; Z; Z^2
rdMp, rdQ, rdth, rdHRnum, rdHR71	M'; $Q(\theta)$; θ; the numerical mixing (60 digits); its values for 7+1

12.2 Names of the earlier sections that section 26 uses

name	section	meaning
X, ID4	1	the coordinates $\{x_0, \ldots, x_7\}$; the 4×4 unit matrix
curvature[], gCos, cvCos, a3[x0], b4[x0]	11	the curvature routine; the cosmological metric and its curvature
toH, Ha, Hb, dHa, dHb	11	rewriting a', a'', b', b'' in terms of H_a, H_b and their derivatives
feStress, bS	20	the stress tensor (ε, p_a, p_b) of a comoving energy $W(a, b)$; the symbol b
rhoK	17.2	ρ_k, the transverse 4×4 matrices
vEW, gFermi, mHb	17.3	v, G_F, m_h
rIIMPl, rIIM8, rIIVp, rIIr4	22.2	M_{Pl}, M_8, $V_\perp$, $r_4(k)$
tbLp, tbHbarCm, tbNmu	25	$\ell_\perp$, $\hbar c$, $N(\mu)$ on T^4
check	0	records a named test in the association checks